AI Problem Design: Student Feedback Classifier

1. Problem Statement

Educational institutions receive feedback from students about their courses, teachers, and learning experiences. Manually reviewing and categorizing this feedback can be time-consuming.

The proposed AI solution will classify student feedback into three categories: Positive, Negative, and Neutral.

The goal is to help teachers and educational institutions understand student opinions more efficiently.

2. Target Users

The primary users are:

- Teachers who want to understand student opinions.
- Educational institutions that collect course feedback.
- Academic coordinators who review student satisfaction.

3. AI Use Case

Task Type: Text Classification

The system will receive a short piece of student feedback and predict its sentiment category.

Input: A text message containing student feedback.

Output: One of three labels: Positive, Negative, or Neutral.

Example:

Input: "The teacher explains difficult concepts in an easy way."

Expected Output: Positive

4. Data Source

The initial dataset will contain a small collection of sample student feedback messages created for testing and demonstration purposes.

The dataset will include text feedback and its corresponding sentiment label.

Each record will contain:

- Feedback text
- Sentiment label

The initial dataset will be manually reviewed to check that the labels are reasonable.

If the project is expanded, a publicly available sentiment-analysis dataset may be used, subject to its license and suitability.

No real student names, contact details, or other personal information will be required.

5. Constraints and Assumptions

- The initial dataset will be small and may not represent all students.
- Feedback may contain spelling mistakes, slang, sarcasm, or mixed sentiments.
- The system may misclassify ambiguous feedback.
- The initial version will support English-language feedback only.
- The system will assist human reviewers rather than replace their judgment.
- The system will not make decisions about teachers or students.
- The quality of predictions will depend on the dataset and classification method.

6. Proposed Approach

The project will begin with a small, labeled dataset.

A simple baseline approach, such as a TF-IDF vectorizer combined with a Logistic Regression classifier, will be used to classify feedback.

The dataset will be divided into training and testing sets. The test set will contain examples that the model did not use during training.

Model performance will be measured using accuracy, precision, recall, and F1-score.

A confusion matrix will also be used to identify which sentiment categories are frequently misclassified.

7. Success Criteria

The initial project will be considered successful if:

1. The system correctly processes valid English feedback messages.
2. The model achieves at least 80% accuracy on a held-out test set.
3. Precision, recall, and F1-score are reported for each sentiment category.
4. The system produces a prediction for each supported input.
5. The system handles empty or invalid input without crashing.
6. The evaluation results and limitations are documented.

The 80% accuracy target is a preliminary goal, not a guaranteed result. If the model does not reach this target, its actual performance and possible improvements will be reported honestly.

8. Evaluation Approach

The dataset will be split into training and testing sets before model training.

The model will be evaluated on the held-out test set using:

- Accuracy: The proportion of correct predictions.
- Precision: How often predictions for a category are correct.
- Recall: How many actual examples of a category the model identifies.
- F1-score: A balance between precision and recall.

A confusion matrix will help identify errors between Positive, Negative, and Neutral categories.

The results will be compared against a simple baseline, such as always predicting the most frequent category.

9. Expected Outcome

The expected outcome is a basic AI system that automatically categorizes student feedback and helps academic staff review opinions more efficiently.

The project will demonstrate a practical application of text classification, dataset preparation, model evaluation, and responsible AI development.

10. Future Improvements

Possible future improvements include:

- Supporting Urdu and Roman Urdu feedback.

- Using a larger and more diverse dataset.
- Developing a simple web interface.
- Allowing users to upload multiple feedback messages.
- Evaluating the system on feedback from different educational contexts.

Conclusion

This project defines a narrow and measurable AI problem: classifying student feedback into Positive, Negative, and Neutral categories.

Its success will be evaluated through a held-out test dataset, classification metrics, and documented limitations.